

From Chatbots to Agents

What you are about to build with — and how to direct it

SMDG Vibeathon 2026 · Opening Session · ~30 minutes

Two years ago, AI answered questions. Today, it does the work.

Chatbot → Agent. You go from typist to supervisor.

CHAT you ask —————▶ it answers
AGENT you brief ▶ it plans ▶ [YOU APPROVE] ▶ it builds ▶ it checks ▶ done

The gate: you approve the plan — before it builds anything.

Step one: a machine that predicts the next word

- Trained on a large slice of the world's text — books, code, documents
- Learn the pattern: after "*invoice*" comes "*number*". At massive scale, patterns start to look like understanding
- That one trick — predict the next token, billions of times — started everything
- 2022–23: ChatGPT introduces the world to it. That chatbot you know is **generation one: a brain with no body**

Since then, three directions at once

- **SENSES** — models that see, hear, make images, music, video
- **THINKING** — models that pause and reason before they answer
- **DOING** — models that act: use tools, run code, finish a task end-to-end

This hackathon lives on the third axis. Doing needed one more piece — next slide.

Why coding cracked first — verifiable means loopable

- Coding is **language** — a model trained on text is at home there
- And code is **checkable**: it runs or it crashes. Compiles or it doesn't
- Checkable → the AI can close the loop and get better each round — no human needed in the middle
- The same trick is now working in **math** — wherever an answer can be checked

ACT ► CHECK ► FIX ► ACT ► CHECK ... until it passes your test

Anything you can check, an agent can learn to do — with a loop.

A model alone can't do anything. Add a harness.

- **Model** = the brain: knowledge, judgment, language
- **Harness** = the body: hands (tools), eyes (reading), memory — and the loop from the last slide
- DeepSeek's one-line formula: **Model + Harness = Agent**
- Chatbot = brain on a desk. Agent = brain with a body, doing a job, reporting back

MODEL (brain) + HARNESS (hands · eyes · memory · loop) = AGENT

DeepSeek Harness — nothing hidden, nothing locked in

- Other harnesses exist — **Claude Code, Codex** — but each is locked to one vendor's models
- DeepSeek Harness is **open and model-agnostic**: DeepSeek, Qwen, GLM — models, tools and skills are all swappable plugins
- **Watch it work**: every step is logged — what it read, which tool it used, why it stopped. No black box
- **Safety rails**: it asks for approval before risky actions, and works inside a sandbox

The difference: open, and not married to any one model.

Treat it like a brilliant, literal-minded new hire

Brilliant: knows a lot, works fast, never sleeps. Literal: does exactly what you said — not what you meant.

- 1 MAKE IT ASK questions first — every one now saves ten minutes later
- 2 APPROVE THE PLAN before it builds anything (it may run very long)
- 3 AGREE WHAT "DONE" MEANS in terms YOU can check — what will you see?
- 4 SET THE RAILS a clock and a budget — fail twice, stop and report

Vague brief in → vague product out. Ambiguity is your enemy, not the AI's.

One agent, or a team? Give it staff.

- Your **main agent** = your single point of contact — the chief of staff
- **Sub-agents** for the heavy lifting: research, coding, writing — each with its own model, focus and memory
- Why split? One brain juggling everything gets distracted. Specialists stay sharp — and you stay in charge
- Beginner rule: **start with one agent**. Add specialists when the job asks for them

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YOU ► MAIN AGENT (your contact, keeps the plan)
      └─► RESEARCH agent (digests your track brief)
      └─► CODING agent   (builds and runs the code)
      └─► WRITING agent  (final summary)
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The future you take home: your models, your call

- The biggest models live in the cloud
- Open models mean AI can also run **where you run** — your machine, your data never leaves
- Agents can call other agents — across cloud and local — you choose which brain, where, and who sees what

CLOUD agents (biggest models) $\rightleftharpoons$ LOCAL agents (open models, your data stays yours)

Same harness you hack with today does both. Sovereignty as a feature — not science fiction.

What happens next

- **5 teams · 3 tracks** — ship planning & schedules · terminal performance & transshipment · data standards & EDI
- Your **track briefing and sample data** are ready — read both before you brief your agent
- Right after this: a short **demo** of the harness
- Then: **your accounts, your team, your clock**

Go build something your ops team would be jealous of.